volatility_filter_legacy

equity strategy | family: equity-volatility | status: reference | derived from `volatility_filter`

What this strategy does

Balanced Leverage, skipping long/short x jumpiness groups that have not made money so far -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Skip the cells that history says do not pay -- but the data shows a gradient, not a bad cell.

Design notes (from the strategy's source file).

Volatility Filter (and Calm Weighted Filter) -- skip stock groups that have not paid, judged only on past data.

Drops a (side, idio-vol tercile) cell from new entries when its own history says it has not earned its keep, judged ONLY on quarters whose positions had fully exited. Result: filtering does not help on its own (corrected run: Sharpe 0.89 vs 1.09 for Balanced Leverage; pre-fix 1.23 vs 1.30; the effect is a smooth gradient, not one broken cell) and adds nothing once inverse-vol sizing is on (corrected: Calm Weighted Filter 1.08 < Calm Weighted 1.24; pre-fix 1.41 < 1.47). Kept as a documented negative result: size by vol, do not filter by it.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	volatility filter legacy	market (CRSP VW)
Annualized return	6.27%	9.94%
Annualized volatility	5.05%	19.35%
Sharpe (NAV-pct)	1.23	0.59
Sortino	1.79	
Max drawdown	-16.7%	-55.5%
Calmar	0.38	
Final NAV ($10M start)	$32,665,477	
Worst day	-3.65%	
% positive days	51.8%	
Sharpe 1996-2001	1.90	
Sharpe 2002-2007	1.44	

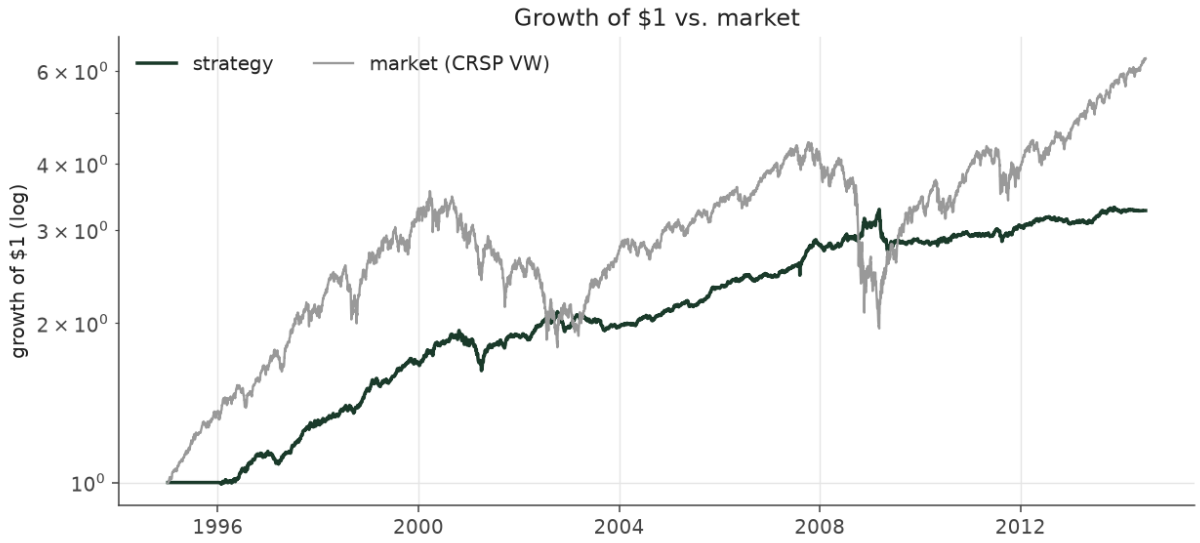
Sharpe 2008-2013	0.48	

Correlation of daily returns with the market: **0.04**.

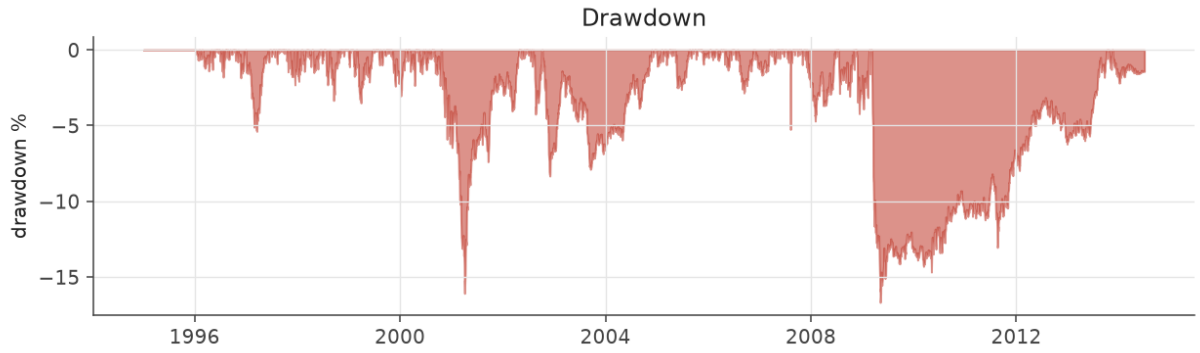
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.23

• best return among equity: `market_boost_max` (0.116); this one: 0.0627

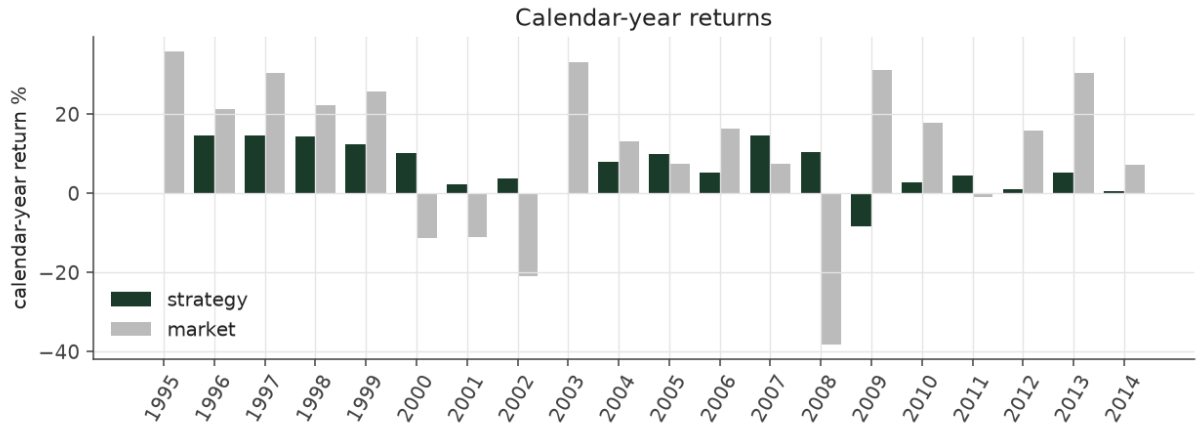
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.376



NAV (log scale)



Drawdown from running peak

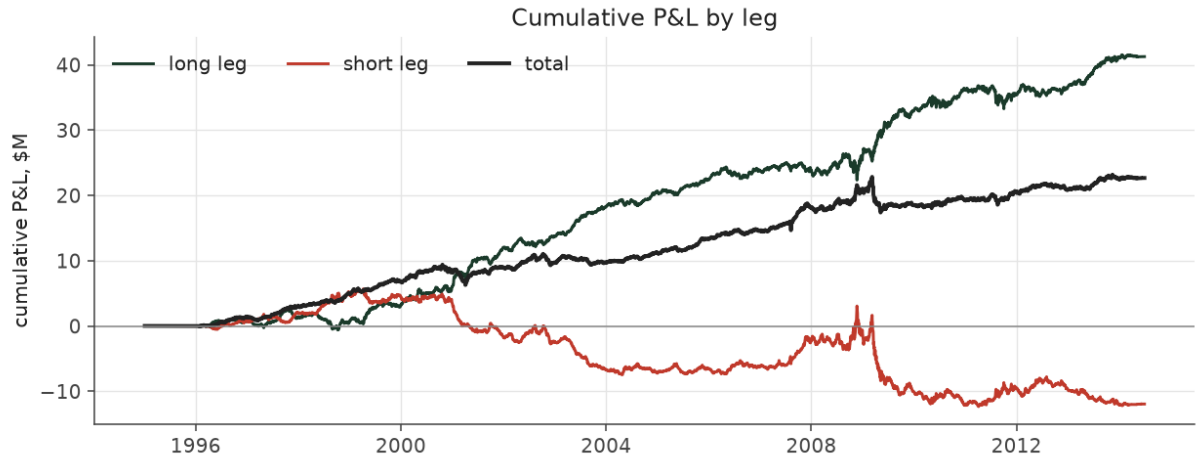


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	41.25	-11.96
P&L after costs ($M)	38.06	-15.40
of which common (average-event) component ($M)	25.51	-26.25

of which excess / PEAD component ($M)	15.28	14.63
excess minus costs ($M)	12.11	11.21
gross return per $ traded (bp)	253.40	-71.65
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.21	-0.26

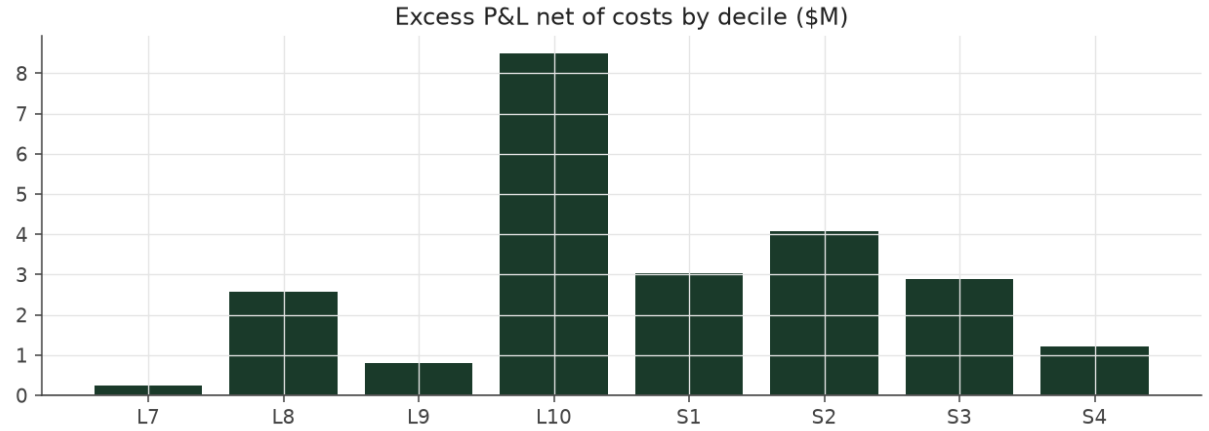
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-12.0M before costs) while the long leg earns +41.3M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 25.5M / 26.3M. Stripping that common component, the surprise signal itself earned +15.3M on the long side and +14.6M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.21 long / -0.26 short vs 1.23 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,009	143.22	2.92	0.52	0.28	0.24	1.0%	16.48
long	8	16,508	381.09	8.90	3.32	0.75	2.56	11.0%	67.22
long	9	17,676	504.10	8.05	1.77	0.97	0.80	3.4%	15.83
long	10	17,111	599.56	18.19	9.68	1.18	8.50	36.4%	141.72
short	1	17,093	602.76	-5.93	4.32	1.29	3.03	13.0%	50.31
short	2	17,204	519.78	-4.61	5.12	1.06	4.07	17.4%	78.22
short	3	15,897	390.13	-3.46	3.67	0.79	2.89	12.4%	73.99
short	4	9,154	156.96	-1.39	1.51	0.30	1.21	5.2%	76.97



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,166	300.07	5.43	7.41	1.15	6.26	26.8%	208.54
2	22,785	620.13	5.83	7.86	1.74	6.12	26.3%	98.72

3	25,396	756.47	5.70	7.07	1.58	5.49	23.5%	72.56
4	27,473	807.13	3.43	4.48	1.26	3.22	13.8%	39.94
5	28,832	813.80	2.29	3.09	0.89	2.19	9.4%	26.96

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,618	221.29	5.14	6.67	1.11	5.56	23.8%	251.21
2	23,010	655.74	5.53	8.33	1.97	6.37	27.3%	97.09
3	25,773	804.72	5.97	7.99	1.61	6.38	27.4%	79.29
4	27,539	811.43	4.28	4.79	1.14	3.65	15.7%	44.98
5	28,707	804.24	1.74	2.13	0.80	1.32	5.7%	16.45

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,693	1,250.38	-0.79	6.28	2.20	4.08	17.5%	32.63
41-60d	25,536	745.51	4.38	4.46	1.53	2.94	12.6%	39.38
61-90d	35,163	970.88	6.82	8.23	2.05	6.18	26.5%	63.66
>90d	12,260	330.83	12.25	10.94	0.84	10.09	43.3%	304.98

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,652	695.59	8.69	10.14	1.36	8.78	37.7%	126.28
2002-07	39,722	1,084.81	9.34	11.50	2.13	9.36	40.2%	86.31
2008-13	45,278	1,517.20	4.64	8.27	3.13	5.14	22.0%	33.87

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,907	322.97	1.41	2.11	0.68	1.43	6.1%	44.12
Chems	2,587	220.11	1.30	1.63	0.39	1.24	5.3%	56.51
Durbl	2,836	237.64	2.34	2.84	0.44	2.41	10.3%	101.27
Enrgy	6,186	269.12	3.79	4.37	0.51	3.87	16.6%	143.66
Hlth	11,622	313.38	2.57	3.21	0.66	2.55	10.9%	81.33
Manuf	12,007	294.12	1.73	2.39	0.61	1.78	7.6%	60.39
Money	17,569	288.48	2.51	3.32	0.65	2.67	11.5%	92.59
NoDur	5,074	265.27	1.25	1.74	0.51	1.23	5.3%	46.34
Other	18,309	313.12	2.26	3.10	0.70	2.41	10.3%	76.87
Shops	11,943	306.44	3.05	3.69	0.63	3.05	13.1%	99.66

Telcm	3,942	260.75	-0.03	0.47	0.47	-0.01	-0.0%	-0.36
Utils	4,670	206.21	0.48	1.05	0.38	0.67	2.9%	32.43

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,046	1,043.79	3.54	6.67	1.89	4.78	20.5%	45.77
low vol	40,091	1,128.50	9.89	12.40	2.49	9.91	42.5%	87.80
mid vol	37,903	1,073.74	9.08	10.74	2.13	8.61	36.9%	80.21

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.27%	0.00%
0.5%	1.7	5.98%	0.29%
1.0%	3.4	5.67%	0.59%
2.0%	6.8	5.01%	1.26%
5.0%	16.9	2.36%	3.90%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.